Hazel Chen

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linkedin.com/in/hazelchens Project Portfolio GitHub

Education

University of Virginia Expected May 2028
B.S. Mechanical Engineering, Minor in Electrical Engineering GPA: 3.64 Charlottesville, VA

Engineering Experience

Virginia Motorsports FSAE Team virginiamotorsportseducation.com Charlottesville, VA
Electrical & High Voltage Lead 2025 to Present

- Lead a 20+ member electrical subteam across battery, PCB, and HV systems, owning tractive system architecture decisions and design reviews
- Oversaw PCB development for 10+ new board designs, personally designing the High Voltage Controller, Segment BMS boards, and HV indicator light
- Designed and built the team's 6.8 kWh, 556 V max tractive battery (128S1P pouch cells, eight vertically stacked 16S modules), owning segment assembly, electronics bay design, and busbar layout
- Assembled 16S modules with aluminum heat spreaders and Poron foam between cells and folded, bolted cell tab series connections with connections directly to cell tap PCB.
- Validated precharge timing and comparator crossover with an oscilloscope and logic analyzer, confirming the main contactor closes at 93% of pack voltage.

Student Researcher, UVA WEARLab Summer 2026

- Developed a portable fNIRS device for monitoring hemoglobin levels in ICU patients to support Takotsubo syndrome research
- Wrote nRF52840 firmware (Arduino C++) to time multiplex IR LED pulses and sample photodiodes, adding dead time between pulses to reduce optical crosstalk

Projects *Full Portfolio*

High Voltage Controller (HVC) PCB for FSAE EV Racecar [Project Page]

- Designed a 2 layer HVC in Altium handling precharge, contactor sequencing, and fail safe discharge for a 556 V pack, consolidating the previous season's multi board system onto one board
- Implemented an AMC0381D isolated amplifier to allow software monitoring of precharge voltage, used in conjunction with an analog comparator circuit to enable precharge
- Integrated an ADBMS6822 dual isoSPI transceiver in a reversible ring topology, allowing redundancy if one link fails
- Fault handling with latched IMD/BMS faults with manual reset, contactor aux contact readback for weld detection, and a mosfet that discharges the HV bus if HVC power is lost

Segment BMS, Rev 2 (In Progress) [Project Page]

- Replaced the Rev 1 stacking connector with an FPC interconnect, freeing space to cut the cell tap board from 6 layers to 2 and improving vibration tolerance and ease of assembly

Battery Thermal Model (MATLAB) [Project Page]

- Built a MATLAB program coupling cell temperature and state of charge over a 20 lap FSAE endurance LapSim power trace for a 140S3P Molicel P50B pack, using only manufacturer published DCIR and OCV vs SoC data
- Predicted a 30 to 42.5 °C cell temperature rise allowing us to inform cooling requirements before building hardware

Battery Pack Fusible Link Testing and Simulation [Project Page]

- Built a custom constant current test rig to fuse dogbone style nickel strip links of varying widths, observing resistance and dissipated power rise together until the link opened
- Ran coupled transient electro-thermal simulations in ANSYS Mechanical at constant current with temperature dependent resistivity to model the runaway seen in testing
- Found the simulation did not match measured results and that its predictions depended heavily on the assumed thermal boundary conditions

Technical Skills *Full Skills List*

Design: Altium Designer, KiCad, Autodesk Fusion (CAD/CAM), ANSYS Mechanical
Programming: C/C++ (Arduino: nRF52840, SAMD21), CircuitPython (RP2040), MATLAB **Protocols:** SPI/isoSPI, I2C, CAN

Embedded Systems & Programming

- Programmed an RP2040 in CircuitPython to control DAC dimmed LED drivers through an OLED and rotary encoder UI
- Brought up a custom SAMD21 board, flashing the bootloader via J Link and integrating I2C sensors

Electronics Test Equipment

- Probed and decoded isoSPI traffic between the segment BMS and MCU with a logic analyzer, validating readings against firmware
- Ran board bring up and troubleshooting on a current limited linear PSU

Autodesk Fusion

- Designed mounts and packaged electronics inside the tractive battery, including contactor mounting, insulation placement, busbar routing, and fuse placement
- Modeled HV cable routing in CAD to verify fit and bend clearance, keeping LV contactor coil wiring separated from HV

Languages: Mandarin Chinese (Fluent), English (Native)